

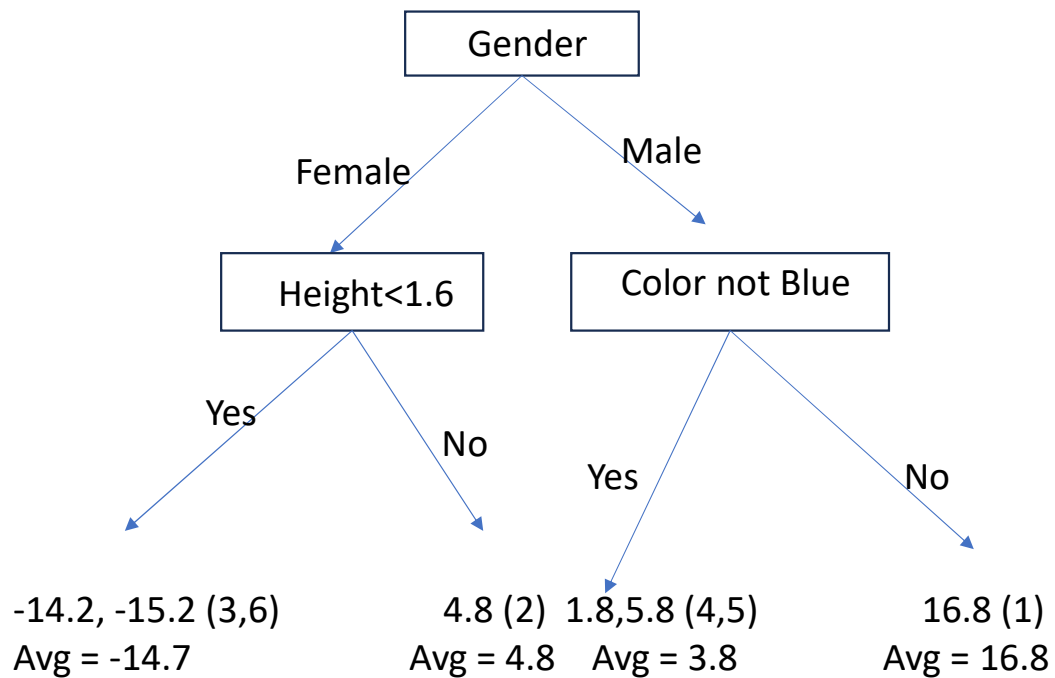
Gradient Boosting for Regression and Classification

Height (m)	Favourite Color	Gender	Weight (kg)	Height (m)	Favourite Color	Gender	Weight (kg)	Residuals
1.6	Blue	Male	88	1.6	Blue	Male	88	16.8
1.6	Green	Female	76	1.6	Green	Female	76	4.8
1.5	Blue	Female	56	1.5	Blue	Female	56	-15.2
1.8	Red	Male	73	1.8	Red	Male	73	1.8
1.5	Green	Male	77	1.5	Green	Male	77	5.8
1.4	Blue	Female	57	1.4	Blue	Female	57	-14.2

Initial Prediction = Average value = 71.2

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Now build a regression tree to predict the residual



Updated prediction for 1st example

$$\begin{aligned}
 &= \text{Initial prediction} + \text{learning rate} \times \text{Predicted by tree} \\
 &= 71.2 + 0.1 \times 16.8 \\
 &= 72.9
 \end{aligned}$$

New residual for 1st example

$$= 88 - 72.9 = 15.1$$

Updated prediction for 2nd example

$$\begin{aligned}
 &= \text{Initial prediction} + \text{learning rate} \times \text{Predicted by tree} \\
 &= 71.2 + 0.1 \times 4.8 \\
 &= 71.68
 \end{aligned}$$

New residual for 2nd example

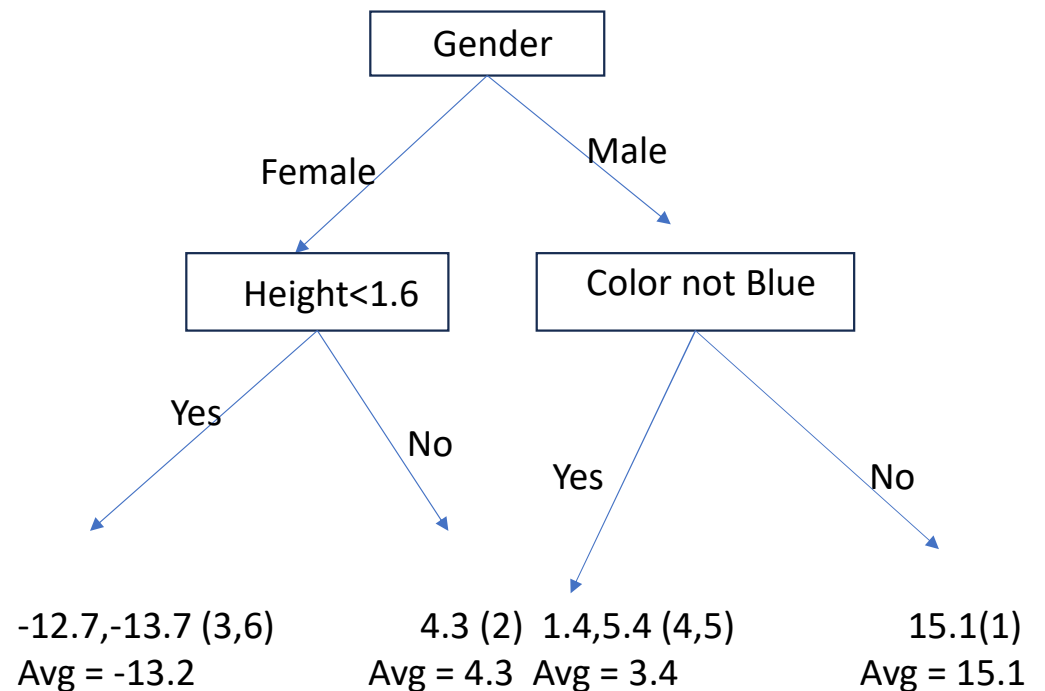
$$= 88 - 71.68 = 16.32$$

Updated prediction for 6th example

$$= 71.2 + 0.1 \times (-14.7) = 69.7$$

$$\text{New residual for 6th example} = 57 - 69.7 = -12.7$$

Height (m)	Favorite Color	Gender	Weight (kg)	Residuals after 1 st iteration
1.6	Blue	Male	88	15.1 (72.9)
1.6	Green	Female	76	4.3 (71.68)
1.5	Blue	Female	56	-13.7 (69.7)
1.8	Red	Male	73	1.4 (71.6)
1.5	Green	Male	77	5.4 (71.6)
1.4	Blue	Female	57	-12.7 (69.7)



Updated prediction for 1st example = First guess + 0.1 x prediction by 1st tree + 0.1 x prediction by 2nd tree
 = 71.2 + 0.1x16.8 + 0.1x15.1
 = 74.4

Updated prediction for 3rd example = First guess + 0.1 x prediction by 1st tree + 0.1 x prediction by 2nd tree
 = 71.2 + 0.1x(-14.7) + 0.1x(-13.2)
 = 68.41

Height (m)	Favorite Color	Gender	Weight (kg)	Residuals after 2nd iteration
1.6	Blue	Male	88	13.6 (74.4)
1.6	Green	Female	76	3.9 (72.1)
1.5	Blue	Female	56	-12.4 (68.4)
1.8	Red	Male	73	1.1 (71.9)
1.5	Green	Male	77	5.1 (71.9)
1.4	Blue	Female	57	-11.4 (68.4)

Now build a 3rd regression tree to predict the latest residuals.

Now prediction of an example = Initial guess + 0.1x prediction by 1st tree + 0.1x prediction by 2nd tree + 0.1x prediction by 3rd tree + so on until the maximum number of iteration is done or the updated residuals become insignificant

Gradient Boost for Classification

Loves Popcorn	Favorite Color	Age	Loves Troll2	Residuals after initial guess
1.6	Blue	12	Yes	0.3
1.6	Green	87	Yes	0.3
1.5	Blue	44	No	-0.7
1.8	Red	19	No	-0.7
1.5	Green	32	Yes	0.3
1.4	Blue	14	Yes	0.3

Log of odds = $\log(4/2) = 0.7$

Probability of loving Troll2 = $1/(1+e^{-0.7}) = 0.7$

Value for each leaf

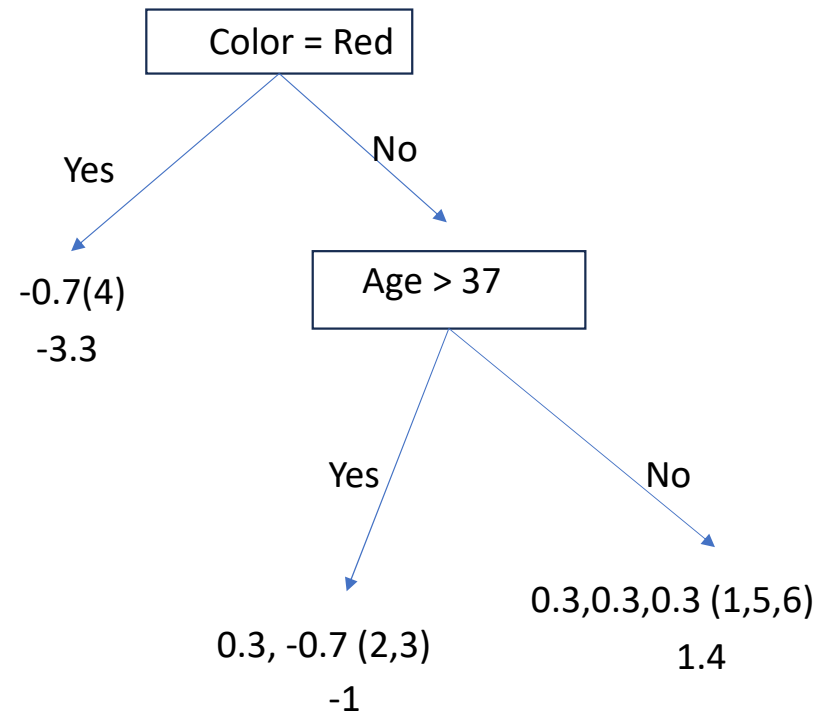
$$= \frac{\text{Sum(Residuals)}}{\text{sum}\{ (\text{Previous probability})(1- \text{Previous probability}) \}}$$

Value for 3rd leaf

$$= \frac{0.3+0.3+0.3}{0.7(1-0.7) + 0.7 (1-0.7) + 0.7 (1-0.7)}$$

= 1.4

Log odds for 3rd example = $0.7 + 0.8 \times (-1) = -0.1$
 Probability pf loving Troll2 = $1/(1+e^{0.1}) = 0.5$



Log odds for 1st example = $0.7 + 0.8 \times 1.4 = 1.8$
 Probability pf loving Troll2 = $1/(1+e^{-1.8}) = 0.9$

Loves Popcorn	Favorite Color	Age	Loves Troll2	Residuals after 1 st iteration	Probability after 1 st iteration
1.6	Blue	12	Yes	0.1	0.9
1.6	Green	87	Yes	0.1	0.9
1.5	Blue	44	No	-0.5	0.5
1.8	Red	19	No	-0.1	0.1
1.5	Green	32	Yes	0.1	0.9
1.4	Blue	14	Yes	0.1	0.9

Value for each leaf

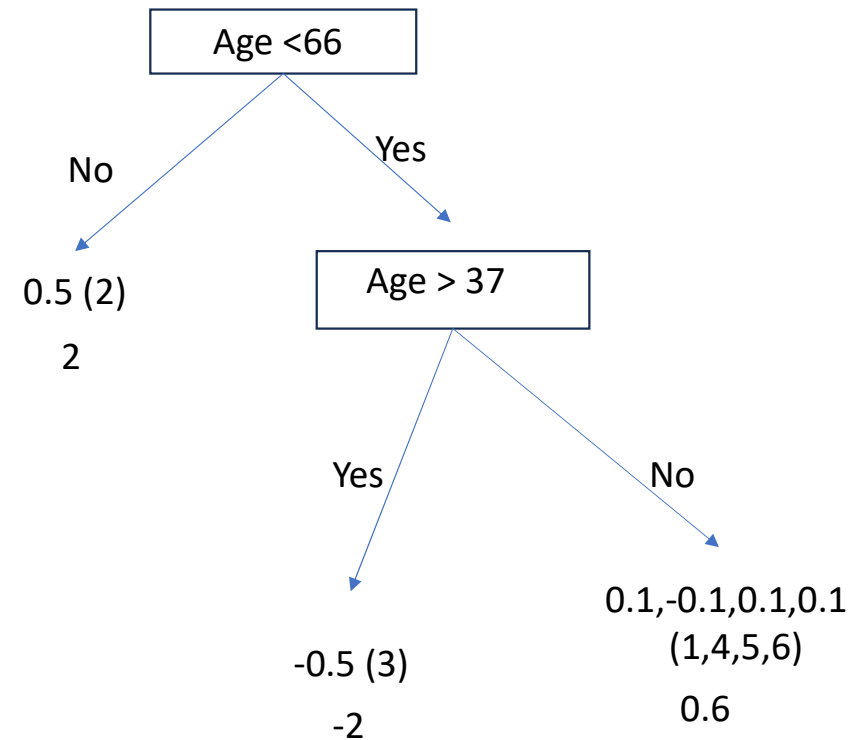
$$= \frac{\text{Sum(Residuals)}}{\text{sum}\{ (\text{Previous probability})(1 - \text{Previous probability}) \}}$$

Value for 3rd leaf

$$= \frac{0.1 - 0.1 + 0.1 + 0.1}{0.9(1 - 0.9) + 0.1(1 - 0.1) + 0.9(1 - 0.9) + 0.9(1 - 0.9)}$$

$$= 0.6$$

Log odds for 3rd example = $0.7 + 0.8 \times (-1) + 0.8 \times (-2) = -1.7$
 Probability pf loving Troll2 = $1/(1 + e^{1.7}) = 0.15$



Log odds for 1st example = $0.7 + 0.8 \times 1.4 + 0.8 \times 0.6 = 2.3$
 Probability of loving Troll2 = $1/(1 + e^{-2.3}) = 0.91$

Loves Popcorn	Favorite Color	Age	Loves Troll2	Residuals after 2nd iteration	Probability after 1 st iteration
1.6	Blue	12	Yes	0.09	0.91
1.6	Green	87	Yes	0.1	0.9
1.5	Blue	44	No	-0.15	0.15
1.8	Red	19	No	-0.1	0.1
1.5	Green	32	Yes	0.1	0.9
1.4	Blue	14	Yes	0.1	0.9